

```
[mysql> use srikar;
```

Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> CREATE TABLE physician (  
->     employeeid INT PRIMARY KEY,  
->     name VARCHAR(100),  
->     position VARCHAR(100),  
->     ssn VARCHAR(20)  
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> INSERT INTO physician VALUES  
-> (1, 'John Dorian', 'Staff Internist', '11111111'),  
-> (2, 'Elliot Reid', 'Attending Physician', '22222222'),  
-> (3, 'Christopher Turk', 'Surgical Attending Physician', '33333333'),  
-> (4, 'Percival Cox', 'Senior Attending Physician', '44444444'),  
-> (5, 'Bob Kelso', 'Head Chief of Medicine', '55555555'),  
-> (6, 'Todd Quinlan', 'Surgical Attending Physician', '66666666'),  
-> (7, 'John Wen', 'Surgical Attending Physician', '77777777'),  
-> (8, 'Keith Dudemeister', 'MD Resident', '88888888'),  
-> (9, 'Molly Clock', 'Attending Psychiatrist', '99999999');
```

Query OK, 9 rows affected (0.01 sec)

Records: 9 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM physician;
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
4	Percival Cox	Senior Attending Physician	444444444
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999

```
+-----+-----+-----+
+
9 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE department (
->   departmentid INT PRIMARY KEY,
->   name VARCHAR(100),
->   head INT
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO department VALUES
-> (1, 'General Medicine', 4),
-> (2, 'Surgery', 7),
-> (3, 'Psychiatry', 9);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM department;
+-----+-----+-----+
| departmentid | name           | head |
+-----+-----+-----+
|          1 | General Medicine |    4 |
|          2 | Surgery         |    7 |
|          3 | Psychiatry      |    9 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT d.name AS Department,
->          p.name AS Head_Physician
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid;
+-----+-----+
| Department      | Head_Physician |
+-----+-----+
| General Medicine | Percival Cox   |
| Surgery         | John Wen       |
| Psychiatry      | Molly Clock    |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM physician
-> WHERE position='Surgical Attending Physician';
```

employeeid	name	position	ssn
3	Christopher Turk	Surgical Attending Physician	333333333
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777

```
3 rows in set (0.00 sec)
```



```
mysql> SELECT *
-> FROM physician
-> WHERE name LIKE 'John%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777

```
2 rows in set (0.00 sec)
```



```
mysql> SELECT COUNT(*) AS Total_Physicians
-> FROM physician;
```

Total_Physicians
9

```
1 row in set (0.00 sec)
```



```
mysql> SELECT COUNT(DISTINCT position)
-> FROM physician;
```

COUNT(DISTINCT position)
7

```
1 row in set (0.00 sec)
```

```

mysql> Welcome to the MySQL monitor.  Commands end with ; or \g.
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'Welcome to the MySQL monitor.  Commands end with' at line 1
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'or' at line 1
-> Your MySQL connection id is 15
-> Server version: 8.0.22 MySQL Community Server - GPL
->
-> Copyright (c) 2000, 2020, Oracle and/or its affiliates. All rights re
served.
->
-> Oracle is a registered trademark of Oracle Corporation and/or its
-> affiliates. Other names may be trademarks of their respective
-> owners.
->
-> Type 'help;' or '\h' for help. Type '\c' to clear the current input s
tatement.
->
-> mysql> USE rakshitha;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '.
Your MySQL connection id is 15
Server version: 8.0.22 MySQL Community Server -' at line 1
mysql> Database changed
-> mysql> CREATE TABLE physician (
->     ->     employeeid INT PRIMARY KEY,
->     ->     name VARCHAR(100),
->     ->     position VARCHAR(100),
->     ->     ssn VARCHAR(20)
->     -> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'Database changed
mysql> CREATE TABLE physician (
->     employeeid INT PRIMA' at line 1
mysql> Query OK, 0 rows affected (0.03 sec)
->

```

```

-> mysql> INSERT INTO physician VALUES
->     -> (1,'John Dorian','Staff Internist','11111111'),
->     -> (2,'Elliot Reid','Attending Physician','22222222'),
->     -> (3,'Christopher Turk','Surgical Attending Physician','33333333
3'),
->     -> (4,'Percival Cox','Senior Attending Physician','4444444444'),
->     -> (5,'Bob Kelso','Head Chief of Medicine','555555555'),
->     -> (6,'Todd Quinlan','Surgical Attending Physician','6666666666'),
->     -> (7,'John Wen','Surgical Attending Physician','777777777'),
->     -> (8,'Keith Dudemeister','MD Resident','888888888'),
->     -> (9,'Molly Clock','Attending Psychiatrist','999999999');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO physician VALUES
->' at line 1
mysql> Query OK, 9 rows affected (0.01 sec)
-> Records: 9 Duplicates: 0 Warnings: 0
->
-> mysql> SELECT * FROM physician;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'Query OK, 9 rows affected (0.01 sec)
Records: 9 Duplicates: 0 Warnings: 0

```

```

mys' at line 1
mysql> +-----+-----+-----+-----+
-----+
-> | employeeid | name                | position                | ssn
-> +-----+-----+-----+-----+
-----+
-> |          1 | John Dorian         | Staff Internist         | 111
111111
-> |          2 | Elliot Reid         | Attending Physician     | 222
222222
-> |          3 | Christopher Turk    | Surgical Attending Physician | 333
333333
-> |          4 | Percival Cox        | Senior Attending Physician | 444
444444
-> |          5 | Bob Kelso           | Head Chief of Medicine   | 555
555555
-> |          6 | Todd Quinlan        | Surgical Attending Physician | 666
666666
-> |          7 | John Wen            | Surgical Attending Physician | 777
777777
-> |          8 | Keith Dudemeister   | MD Resident             | 888
888888
-> |          9 | Molly Clock         | Attending Psychiatrist   | 999
999999
-> +-----+-----+-----+-----+
-----+
-> 9 rows in set (0.00 sec)
->
-> mysql> CREATE TABLE department (
->     ->     departmentid INT PRIMARY KEY,
->     ->     name VARCHAR(100),
->     ->     head INT
->     -> );

```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '+'-----+-----+-----+-----+-----+-----+-----+-----+-----+
---+
```

```
| ' at line 1
mysql> Query OK, 0 rows affected (0.02 sec)
```

```
->
-> mysql> INSERT INTO department VALUES
->      -> (1,'General Medicine',4),
->      -> (2,'Surgery',7),
->      -> (3,'Psychiatry',9);
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO department VALUES
    -' at line 1
mysql> Query OK, 3 rows affected (0.01 sec)
    -> Records: 3 Duplicates: 0 Warnings: 0
    ->
```

```
-> mysql> SELECT * FROM department;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mys' at line 1
```

```
mysql> +-----+
-> | departmentid | name | head |
-> +-----+
-> | 1 | General Medicine | 4 |
-> | 2 | Surgery | 7 |
-> | 3 | Psychiatry | 9 |
-> +-----+
-> 3 rows in set (0.00 sec)
->
-> mysql> SELECT d.name AS Department,
->          p.name AS Head_Physician
->          FROM department d
->          JOIN physician p
```

```

->      -> ON d.head = p.employeeid;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '+' at line 1
| departmentid | name | h' at line 1
mysql> +-----+-----+
-> | Department | Head_Physician |
-> +-----+-----+
-> | General Medicine | Percival Cox |
-> | Surgery | John Wen |
-> | Psychiatry | Molly Clock |
-> +-----+-----+
-> 3 rows in set (0.00 sec)
->
-> mysql> SELECT *
->      -> FROM physician
->      -> WHERE position='Surgical Attending Physician';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '+' at line 1
| Department | Head_Physician |
+---' at line 1
mysql> +-----+-----+-----+-----+
-----+
-> | employeeid | name | position | ssn
|
-> +-----+-----+-----+-----+
-----+
-> | 3 | Christopher Turk | Surgical Attending Physician | 3333
33333 |
-> | 6 | Todd Quinlan | Surgical Attending Physician | 6666
66666 |
-> | 7 | John Wen | Surgical Attending Physician | 7777
77777 |
-> +-----+-----+-----+-----+
-----+
-> 3 rows in set (0.00 sec)
->
-> mysql> SELECT *
->      -> FROM physician
->      -> WHERE name LIKE 'John%';

```



```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '-----+-----+-----+-----+
--+
| e' at line 1
mysql> -----+-----+-----+-----+
-+
-> | employeeid | name          | position          | ssn
|
-> +-----+-----+-----+-----+
-+
-> |          1 | John Dorian | Staff Internist   | 111111111
|
-> |          7 | John Wen    | Surgical Attending Physician | 777777777
|
-> +-----+-----+-----+-----+
-+
-> 2 rows in set (0.00 sec)
->
-> mysql> SELECT COUNT(*) AS Total_Physicians
->      -> FROM physician;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '-----+-----+-----+-----+
| employ' at line 1
mysql> -----+
-> | Total_Physicians |
-> +-----+
-> |          9 |
-> +-----+
-> 1 row in set (0.00 sec)
->
-> mysql> SELECT COUNT(DISTINCT position)
->      -> FROM physician;

```

FROM physician;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '+'

```
| Total_Physicians |
+-----+
| ' at line 1
```

```
mysql> +-----+
-> | COUNT(DISTINCT position) |
-> +-----+
-> | 7 |
-> +-----+
-> 1 row in set (0.00 sec)
->
-> ^C
```

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position;
```

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.00 sec)

```
mysql> SELECT position,
->      COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> ORDER BY name ASC;
```

employeeid	name	position	ssn
5	Bob Kelso	Head Chief of Medicine	555555555
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666

9 rows in set (0.00 sec)

```
mysql> SELECT *  
-> FROM physician  
-> ORDER BY employeeid DESC;
```

employeeid	name	position	ssn
9	Molly Clock	Attending Psychiatrist	999999999
8	Keith Dudemeister	MD Resident	888888888
7	John Wen	Surgical Attending Physician	777777777
6	Todd Quinlan	Surgical Attending Physician	666666666
5	Bob Kelso	Head Chief of Medicine	555555555
4	Percival Cox	Senior Attending Physician	444444444
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111

9 rows in set (0.00 sec)

```
mysql> SELECT *  
-> FROM physician  
-> WHERE employeeid IN  
-> (  
->     SELECT head  
->     FROM department  
-> );
```

employeeid	name	position	ssn
4	Percival Cox	Senior Attending Physician	444444444
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

3 rows in set (0.00 sec)

```
mysql> SELECT *  
-> FROM physician  
-> WHERE employeeid NOT IN  
-> (  
->     SELECT head  
->     FROM department  
-> );
```

```

+-----+-----+-----+-----+
+ | employeeid | name           | position           | ssn               |
+-----+-----+-----+-----+
+ | 1 | John Dorian   | Staff Internist    | 111111111         |
+ | 2 | Elliot Reid   | Attending Physician | 222222222         |
+ | 3 | Christopher Turk | Surgical Attending Physician | 333333333         |
+ | 5 | Bob Kelso     | Head Chief of Medicine | 555555555         |
+ | 6 | Todd Quinlan  | Surgical Attending Physician | 666666666         |
+ | 8 | Keith Dudemeister | MD Resident       | 888888888         |
+-----+-----+-----+-----+
+
6 rows in set (0.00 sec)

```

```
mysql> SELECT p.name
-> FROM physician p
-> JOIN department d
-> ON p.employeeid = d.head
-> WHERE d.name = 'Surgery';
```

```
+-----+
| name   |
+-----+
| John Wen |
+-----+
```

1 row in set (0.00 sec)

```
mysql> SELECT d.name,
-> p.name
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid
-> ORDER BY d.name;
```

```
+-----+-----+
| name           | name           |
+-----+-----+
| General Medicine | Percival Cox   |
| Psychiatry       | Molly Clock    |
| Surgery          | John Wen       |
+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE position NOT LIKE '%Surgical%';
```

```
+-----+-----+-----+-----+
| employeeid | name           | position                               | ssN      |
+-----+-----+-----+-----+
| 1          | John Dorian    | Staff Internist                       | 11111111 |
| 2          | Elliot Reid    | Attending Physician                   | 22222222 |
| 4          | Percival Cox   | Senior Attending Physician           | 44444444 |
| 5          | Bob Kelso      | Head Chief of Medicine                | 55555555 |
| 8          | Keith Dudemeister | MD Resident                           | 88888888 |
| 9          | Molly Clock    | Attending Psychiatrist                 | 99999999 |
+-----+-----+-----+-----+
```

6 rows in set (0.00 sec)

```
mysql> SELECT position,
->      COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> ORDER BY Total_Employees DESC;
```

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE position LIKE '%Attending%';
```

employeeid	name	position	ssn
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

6 rows in set (0.00 sec)

```
mysql> SELECT position,
->         COUNT(*) AS Total
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) >= 2;
```

position	Total
Surgical Attending Physician	3

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM department d
-> WHERE EXISTS
-> (
->     SELECT 1
->     FROM physician p
->     WHERE p.employeeid = d.head
-> );
```

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

3 rows in set (0.00 sec)


```
mysql> SELECT MAX(employeeid)
-> FROM physician;
```

MAX(employeeid)
9

```
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(employeeid)
-> FROM physician;
```

MIN(employeeid)
1

```
1 row in set (0.00 sec)
```

```
mysql>
```